



St George Girls High School

2021 TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- For questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks: 100

Section I – 10 marks (pages 2–5)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 6–13)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

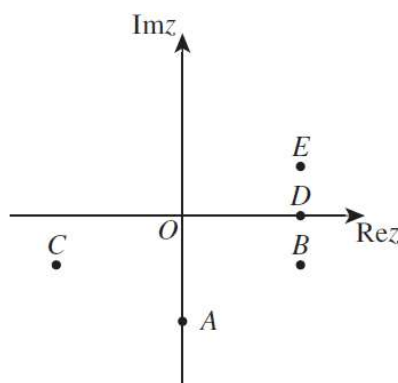
10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Consider the argand diagram, where $z = a + ib$.



Which of the following pairs of points in the complex plane could represent the square roots of z ?

- A. A and D
 - B. B and C
 - C. B and E
 - D. C and E
- 2 Which expression is equal to $\int x e^{\frac{x}{2}} dx$?

- A. $\frac{1}{2} x e^{\frac{x}{2}} - \frac{1}{4} e^{\frac{x}{2}} + C$
- B. $\frac{1}{2} x e^{\frac{x}{2}} - \frac{1}{2} e^{\frac{x}{2}} + C$
- C. $2 x e^{\frac{x}{2}} - 2 e^{\frac{x}{2}} + C$
- D. $2 x e^{\frac{x}{2}} - 4 e^{\frac{x}{2}} + C$

- 3 Which of the following is a valid counter-example to the claim:

‘If it can purr or roar, it’s a cat.’

- A. A cat that can purr but not roar.
 - B. A cat that can’t purr or roar.
 - C. A dog that can roar.
 - D. A dog that can’t purr or roar.
- 4 A sphere has equation $(x - k)^2 + (y - 2k)^2 + (z - 3k)^2 = 49$.

For what value of k does the sphere pass through the origin?

- A. 7
 - B. $\sqrt{7}$
 - C. $\frac{\sqrt{7}}{2}$
 - D. $\sqrt{\frac{7}{2}}$
- 5 Given that a and b are positive real numbers, and $a + b = 4$, find the minimum value of $\left(1 + \frac{1}{a}\right)\left(1 + \frac{1}{b}\right)$.

- A. 0
- B. $\frac{3}{2}$
- C. 4
- D. $\frac{9}{4}$

- 6 For a certain complex number z , it is known that $|z| = 1$.

Which of the following statements must also be true?

- A. $z^{-1} = \bar{z}$
- B. $\text{Arg}(z^{-1}) = \text{Arg}(\bar{z})$
- C. $|z^{-1}| > |\bar{z}|$
- D. $z^{-1} = i\bar{z}$

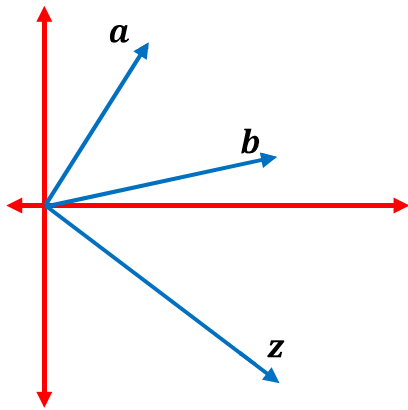
- 7 Which of the following is the Cartesian equation of the line joining $(1, -1, 2)$ and $(-1, 1, 0)$?

- A. $x = -y = z - 1$
- B. $x - 1 = y + 1 = z - 2$
- C. $\frac{x+1}{2} = \frac{y-1}{2} = \frac{z-2}{2}$
- D. $\mathbf{r} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} + \lambda \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$

- 8 Which of the following is a FALSE statement, given $x, y \in \mathbb{Z}$?

- A. $\forall x \exists y (x - y = 0)$
- B. $\forall x \exists y (3x - y = 0)$
- C. $\forall x \exists y (x - 3y = 0)$
- D. $\exists x \exists y (x + y = 8)$

- 9 The vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{z}$ are shown below.



Which of the following could be equivalent to $\mathbf{z}$, where $\lambda_1, \lambda_2 \in \mathbb{R}$?

- A. $\lambda_1 \mathbf{a} + \lambda_2 \mathbf{b}$
 - B. $\mathbf{a} + \lambda_2 \mathbf{b}$
 - C. $\lambda_1 \mathbf{a} + \mathbf{b}$
 - D. None of the above
- 10 How many distinct roots has the equation $(z^4 - 1)(z^2 - 3iz - 2) = 0$, where $z \in \mathbb{C}$?
- A. 3
 - B. 4
 - C. 5
 - D. 6

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

(a) Consider the vectors $\mathbf{u} = 4\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{v} = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$.

- (i) Find the vector $\mathbf{w} = \mathbf{u} + \mathbf{v}$. 1
- (ii) Find the exact value of $|\mathbf{u}|$. 1
- (iii) Hence find the vector $\mathbf{t}$ with magnitude 4, in the direction opposite to $\mathbf{u}$. 1

(b) (i) Find values of a and b such that 2

$$\frac{9x - 2}{(2x + 1)(x - 6)} = \frac{a}{(2x + 1)} + \frac{b}{(x - 6)}.$$

(ii) Hence or otherwise find 2

$$\int \frac{9x - 2}{(2x + 1)(x - 6)} dx.$$

(c) Consider the complex number $z = 1 - \sqrt{3}i$

- (i) Evaluate z^{-1} . 1
- (ii) Express z in the form $e^{i\theta}$. 2
- (iii) Hence, or otherwise, evaluate z^9 . 1
- (iv) Find $\ln z$, in the form $a + ib$, where $a, b \in \mathbb{R}$. 2

Question 11 (continued)

(d) Consider the proposition:

I won't buy your fruit if it's not sweet.

- | | |
|--|----------|
| (i) State the contrapositive of the proposition. | 1 |
| (ii) State a possible counterexample to the proposition. | 1 |

End of Question 11

Question 12 (15 marks)

- (a) (i) Find the square root(s) of $7 - 24i$, in the form $x + iy$, where $x, y \in \mathbb{R}$. **3**
- (ii) Hence or otherwise solve $2z^2 + 6z + (1 + 12i) = 0$, for $z \in \mathbb{C}$. **2**
- (b) Find the angle between $\mathbf{u} = \mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ and $\mathbf{v} = \mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$, correct to the nearest degree. **2**
- (c) (i) Show that $\omega = 2i$ is a solution to $\omega^6 = -64$. **1**
- (ii) Hence find the other distinct solutions to $\omega^6 = -64$, where $\omega \in \mathbb{C}$. **2**
- (d) Use proof by contraposition to prove that if $3k + 1$ is even, then k is odd, $\forall k \in \mathbb{Z}$ **2**
- (e) Find the exact value of $\int_0^1 \sin^{-1} x \, dx$. **3**

End of Question 12

Question 13 (15 marks)

(a) Use proof by contradiction to prove that $\log_6 12$ is irrational. **3**

(b) Find $\int \frac{x^2}{x^2 + 2x + 10} dx$. **3**

(c) A subset of the complex plane is defined by the relation $|z - (4 - 3i)| \leq 3$.

(i) Draw a sketch of this relation. **1**

(ii) Given that z is a complex number that satisfies the relation, find the minimum and maximum values of $|z|$. **2**

(iii) Given that z is a complex number that satisfies the relation, find the minimum and maximum values of $\text{Arg}(z)$. **2**

(d) A curve is defined parametrically as $\mathbf{r} = (\cos t)\mathbf{i} + \mathbf{j} + (\sin^2 t)\mathbf{k}$, for parameter t , $0 \leq t \leq 2\pi$.

(i) Explain why this curve is two-dimensional. **1**

(ii) Sketch the curve, clearly labelling all important features, including the coordinate axes. **3**

End of Question 13

Question 14 (15 marks)

(a) It can be shown that $\frac{a+b}{2} \geq \sqrt{ab}$. DO NOT PROVE THIS.

(i) Prove that $(a+b)(b+c)(c+a) \geq 8abc$ for all positive real numbers a, b , and c . **1**

(ii) Suppose that x, y , and z are the lengths of the sides of a triangle. Using the result from part (i), deduce that **2**

$$xyz \geq (y+z-x)(z+x-y)(x+y-z)$$

(b) (i) Find the constants A and B such that **2**

$$\frac{1}{\cos x} = \frac{A \cos x}{1 - \sin x} + \frac{B \cos x}{1 + \sin x}$$

(ii) Hence find the exact value of $\int_0^{\frac{\pi}{6}} \sec x \, dx$. **3**

(c) Prove that the following three vectors CANNOT be the three sides of a triangle: **2**

$$\mathbf{t} = 4\mathbf{i} - \mathbf{k} - 2\mathbf{j}, \mathbf{u} = 5\mathbf{i} - 2\mathbf{j} + 7\mathbf{k}, \text{ and } \mathbf{v} = \mathbf{i} + 3\mathbf{k} - 2\mathbf{j}$$

(d) It can be shown that $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$. **3**

DO NOT PROVE THIS.

Use this property to find the exact value of $\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} \, dx$.

Question 14 (continued)

- (e) Jimin claims that for the function $f(n) = n^2 + n + 41$, $f(n)$ is prime for all positive integers n .

For example, $f(1) = 43, f(2) = 47, f(3) = 53$ and so on.
DO NOT PROVE THIS.

- (i) Jungkook attempts to disprove this claim by providing the following counter-example: **1**

“7 is a prime number, but 7 isn’t generated by your formula.”

Explain why this is not a valid counter-example.

- (ii) State a counter-example that disproves Jimin’s claim. **1**

End of Question 14

Question 15 (15 marks)

(a) Let $I_n = \int x(\ln x)^n dx$.

(i) Show that $I_n = \frac{1}{2}x^2(\ln x)^n - \frac{1}{2}nI_{n-1}$. **2**

(ii) Hence evaluate $\int_1^3 x(\ln x)^2 dx$. **3**

(b) The lines l_1 and l_2 have vector equations as follows:

$$l_1 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda_1 \begin{pmatrix} 2 \\ 2 \\ -3 \end{pmatrix} \quad l_2 = \begin{pmatrix} 4 \\ -2 \\ 9 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}$$

(i) Show that l_1 and l_2 DO NOT intersect. **3**

(ii) It can be shown that **4**
 $a^2 + b^2 + c^2 \equiv (a + b + c)^2 - 2(ab + ac + bc)$
for $a, b, c \in \mathbb{R}$. DO NOT PROVE THIS.

Using this identity, or otherwise, show that the distance d between a point on l_1 and a point on l_2 is given by

$$d = \sqrt{(3\lambda_2 - 4\lambda_1 - 5)^2 + (\lambda_1 - 1)^2 + 36}$$

(iii) Hence determine the minimum distance between l_1 and l_2 . **1**

(iv) Find the coordinates of the points on the two lines that are the minimum distance apart. **2**

End of Question 15

Question 16 (15 marks)

- (a) Prove the following statement for all r , where $r \in \mathbb{R}$: **2**

If r is irrational, then $\sqrt{r}$ is irrational.

- (b) By evaluation or otherwise, prove that i^i is a real number. **2**

- (c) Let $t = \tan \frac{x}{2}$.

- (i) Show that $\frac{dx}{dt} = \frac{2}{1+t^2}$. **1**

- (ii) Show that $\frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} = \sin x$. **2**

- (iii) Hence, show that $\int_0^{\frac{\pi}{2}} \frac{1}{1+k \sin x} dx = \frac{2}{\sqrt{1-k^2}} \tan^{-1} \sqrt{\frac{1-k}{1+k}}$ **4**
where $0 < k < 1$.

- (iv) Let $I_n = \int_0^{\frac{\pi}{2}} \frac{\sin^n x}{2 + \sin x} dx$, where $n = 0, 1, 2, \dots$ **1**

Show that $I_{n+1} + 2I_n = \int_0^{\frac{\pi}{2}} \sin^n x dx$.

- (v) Hence, or otherwise, find the value of I_2 . Give your answer in the form $m\pi + 1$, where m is irrational. **4**

End of paper

Multiple Choice

① Roots evenly spaced $\therefore C \& E \therefore \textcircled{D}$

② $\int x e^{\frac{x}{2}} dx$ $u = x$
 $u' = 1$ $v = 2e^{\frac{x}{2}}$
 $v' = e^{\frac{x}{2}}$

$$= uv - \int u'v dx$$

$$= 2x e^{\frac{x}{2}} - \int 2e^{\frac{x}{2}} dx$$

$$= 2x e^{\frac{x}{2}} - 4e^{\frac{x}{2}} + C \therefore \textcircled{D}$$

③ $\textcircled{C}$

④ $(k)^2 + (2k)^2 + (3k)^2 = 49$

$$14k^2 = 49$$

$$k^2 = \frac{49}{14}$$

$$= \frac{7}{2}$$

$$k = \sqrt{\frac{7}{2}}$$

$\textcircled{D}$

⑤ Minimum of $(1+\frac{1}{a})(1+\frac{1}{b})$ when neither a nor b are very small

Since $a+b=4$, try $a=b=2$

$$(1+\frac{1}{2})(1+\frac{1}{2}) = \frac{3}{2} \times \frac{3}{2} = \frac{9}{4} \therefore \textcircled{D}$$

or

⑥ $z^{-1} = \frac{1}{z}$

$$= \frac{\bar{z}}{z\bar{z}}$$

$$= \frac{\bar{z}}{|z|^2}$$

$$= \bar{z} \quad (\text{since } |z|=1) \therefore \textcircled{A}$$

⑦ $d = \begin{pmatrix} 2 \\ -2 \\ 2 \end{pmatrix}$ or $\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$

$$\therefore \frac{x-a}{1} = \frac{y-b}{-1} = \frac{z-c}{1}$$

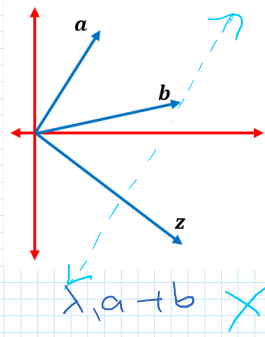
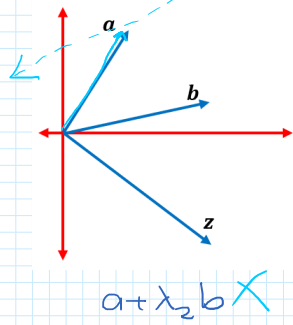
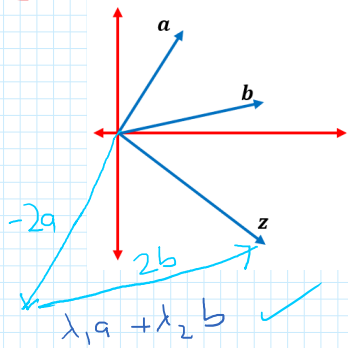
passes through $(0,0,1)$ so

$$\frac{x-0}{1} = \frac{y-0}{-1} = \frac{z-1}{1}$$

$$\therefore x = -y = z-1 \quad \textcircled{A}$$

⑧ $\textcircled{C}$ Eq. for $x=1$, $1-3y=0$,

9



A

10

$$(z^4 - 1)(z^2 - 3(z - 2)) = 0$$

$$(z^2 - 1)(z^2 + 1)(z - 2)(z - 1) = 0$$

$$(z - 1)^2(z + 1)(z - 2)(z^2 + 1) = 0$$

$\therefore 5$ distinct roots
 $\therefore c$

Question 11

a) i $\underline{u} = 4\underline{i} - 2\underline{j} + 3\underline{k} + \underline{i} + 2\underline{j} - 2\underline{k}$
 $= 5\underline{i} + \underline{k}$

ii $|\underline{u}| = \sqrt{4^2 + 2^2 + 3^2}$
 $= \sqrt{16 + 4 + 9}$
 $= \sqrt{29}$

iii $\underline{t} = -4 \times \underline{u}$
 $= \frac{-4}{29} (4\underline{i} - 2\underline{j} + 3\underline{k})$

$$\frac{1}{(2x+1)(x-6)} + \frac{4}{(x-6)} = \frac{4x-6+8x+4}{(2x+1)(x-6)}$$

$$= \frac{12x-2}{(2x+1)(x-6)}$$

b) i $\frac{9x-2}{(2x+1)(x-6)} = \frac{a}{2x+1} + \frac{b}{x-6}$

$$9x-2 = a(x-6) + b(2x+1)$$

when $x=6$, $52=13b$
 $b=4$

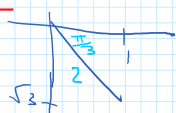
when $b=4$, $a=1$

ii $\int \frac{9x-2}{(2x+1)(x-6)} dx = \int \frac{1}{2x+1} + \frac{4}{x-6} dx$
 $= \frac{1}{2} \ln|2x+1| + 4 \ln|x-6| + C$

c) i $z^{-1} = \frac{1}{1-\sqrt{3}i}$

$$= \frac{1+\sqrt{3}i}{4}$$

$$= \frac{1}{4} + \frac{\sqrt{3}}{4}i$$

ii  $z = 2e^{i\frac{\pi}{3}}$

iii $z^9 = (2e^{i\frac{\pi}{3}})^9$
 $= 512 e^{i\frac{9\pi}{3}}$
 $= 512 e^{i3\pi}$
 $= -512$

iv $\ln z = \ln(2e^{i\frac{\pi}{3}})$
 $= \ln 2 + \ln e^{i\frac{\pi}{3}}$
 $= \ln 2 + i\frac{\pi}{3}$

d) i If I buy your fruit, it's sweet.

ii Buying fruit that's not sweet.

Question 12

a) i let $x+iy = \sqrt{7-24i}$
 $(x+iy)^2 = 7-24i$
 $x^2 - y^2 + 2xyi = 7-24i$
 $\therefore x^2 - y^2 = 7$ $2xy = -24$
 $xy = -12$
 $\therefore x=4, y=-3$ or $x=-4, y=3$
 $\therefore 4-3i$ and $-4+3i$

ii $2z^2 + 6z + (1+12i) = 0$

$$\Delta = b^2 - 4ac$$

$$= 36 - 4 \times 2 \times (1+12i)$$

$$= 36 - 8 - 96i$$

$$= 28 - 96i$$

$$= 4(7-24i)$$

$$\therefore z = \frac{-6 \pm \sqrt{4(7-24i)}}{4}$$

$$= \frac{-6 \pm 2(4-3i)}{4}$$

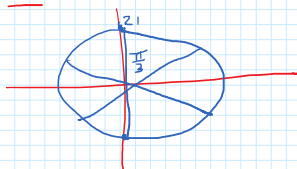
$$= \frac{-3 \pm (4-3i)}{2}$$

$$= \frac{1-3i}{2}, \frac{-7+3i}{2}$$

b) i $u \cdot v = |u||v|\cos\theta$
 $\therefore \cos\theta = \frac{u \cdot v}{|u||v|}$
 $= \frac{1-4-9}{|\sqrt{14}||\sqrt{14}|}$
 $= \frac{-12}{14}$
 $\therefore \theta = 148.997^\circ$
 $\approx 149^\circ$

c) i $(2i)^6 = 2^6 i^6$
 $= 64 \times -1$
 $= -64$

ii $\therefore -2i, 2\text{cis}\frac{\pi}{6}, 2\text{cis}(\frac{\pi}{6}), 2\text{cis}\frac{5\pi}{6}, 2\text{cis}(\frac{5\pi}{6})$



d) i If $3k+1$ is even, then k is odd

contrapositive: If k is even, then $3k+1$ is odd

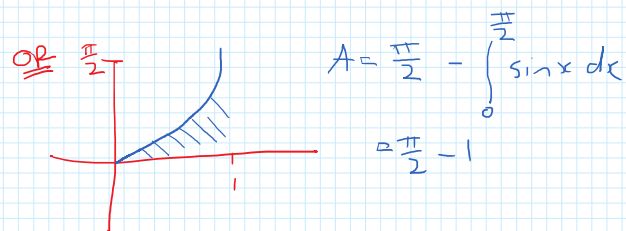
let $k=2a$, $a \in \mathbb{Z}$

$$\begin{aligned}\therefore 3k+1 &= 3(2a)+1 \\ &= 6a+1 \\ &= 2(3a)+1\end{aligned}$$

which is odd, since $2(3a)$ is even ($a \in \mathbb{Z}$)

②

$$\begin{aligned}\int_0^1 \sin^{-1}x \, dx &= \left[x \sin^{-1}x \right]_0^1 - \int_0^1 \frac{x}{\sqrt{1-x^2}} \, dx \quad \begin{array}{l} u = \sin^{-1}x \quad v = x \\ u' = \frac{1}{\sqrt{1-x^2}} \quad v' = 1 \end{array} \\ &= 1 \sin^{-1}(1) + \frac{1}{2} \int_0^1 2x (1-x^2)^{-\frac{1}{2}} \, dx \\ &= \frac{\pi}{2} - \frac{1}{2} \left[2 \sqrt{1-x^2} \right]_0^1 \\ &= \frac{\pi}{2} - [0 - 1] \\ &= \frac{\pi}{2} - 1\end{aligned}$$



Question 13

a) Assume $\log_6 12$ is rational

$\therefore \log_6 12 = \frac{a}{b}$ where $a, b \in \mathbb{Z}^+$ and a, b have no common factors other than one.

$$6^{\frac{a}{b}} = 12$$

$$6^a = 12^b$$

$$(2 \times 3)^a = (2 \times 2 \times 3)^b$$

$$2^a \times 3^a = 2^{2b} \times 3^b$$

$$\therefore a = 2b \text{ and } a = b$$

which is a contradiction

$\therefore \log_6 12$ is irrational.

b)
$$\int \frac{x^2}{x^2+2x+10} dx = \int \frac{x^2+2x+10}{x^2+2x+10} - \frac{2x+2}{x^2+2x+10} - \frac{8}{x^2+2x+10} dx$$

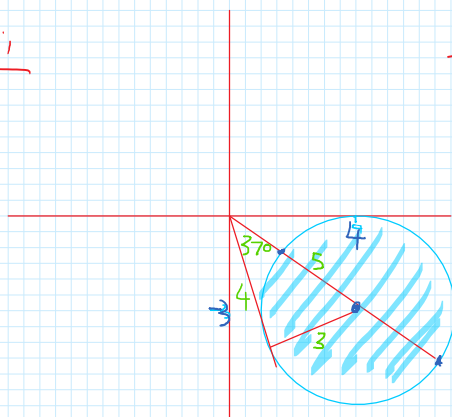
$$= x - \ln|x^2+2x+10| - 8 \int \frac{1}{x^2+2x+10} dx$$

$$= x - \ln|x^2+2x+10| - 8 \int \frac{1}{(x+1)^2+3^2} dx$$

$$= x - \ln|x^2+2x+10| - \frac{8}{3} \tan^{-1} \frac{x+1}{3} + C$$

c)

i

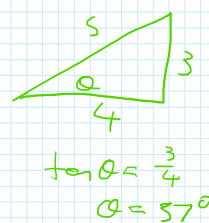


ii $\max |z| = 5+3 = 8$

$\min |z| = 5-3 = 2$

iii $\max \arg(z) = 0$

$\min \arg(z) = 2+37^\circ = 74^\circ$



d) i The y -component is always 1, so the curve exists in the plane $y=1$.

ii $x = \cos t$

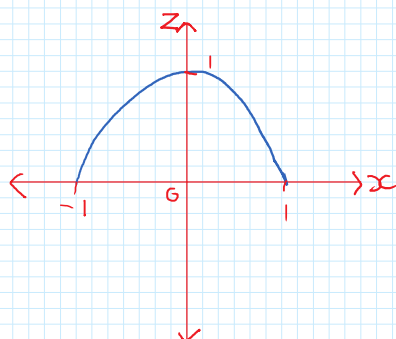
$$z = \sin^2 t$$

$$= 1 - \cos^2 t$$

$$= 1 - x^2$$

since $0 \leq t \leq 2\pi$

and $x = \cos t$,
 $-1 \leq x \leq 1$



Question 14

a) i $a+b > 2\sqrt{ab}$
 $b+c > 2\sqrt{bc}$
 $c+a > 2\sqrt{ac}$

$$\begin{aligned}\therefore (a+b)(b+c)(a+c) &> 2\sqrt{ab} \times 2\sqrt{bc} \times 2\sqrt{ac} \\ &= 8\sqrt{ab \times bc \times ac} \\ &= 8\sqrt{a^2 b^2 c^2} \\ &= 8abc\end{aligned}$$

ii

$$\begin{array}{llll} \text{let } a+b=x & \therefore b=x-a & \Rightarrow b=x-(z-c) & \Rightarrow b=x-(z-(y-b)) \\ b+c=y & c=y-b & c=y-(x-a) & c=y-(x-(z-c)) \\ c+a=z & a=z-c & a=z-(y-b) & a=z-(y-(x-a)) \end{array}$$

$$\begin{aligned}\Rightarrow 2b &= x-z+y \\ 2c &= y-x+z \\ 2a &= z-y+x\end{aligned}$$

$$\begin{aligned}\Rightarrow b &= \frac{x+y-z}{2} \\ c &= \frac{z+y-x}{2} \\ a &= \frac{x+z-y}{2}\end{aligned}$$

Since $(a+b)(b+c)(c+a) > 8abc$

then $(x)(y)(z) > 8\left(\frac{x+z-y}{2}\right)\left(\frac{x+y-z}{2}\right)\left(\frac{z+y-x}{2}\right)$
 $\therefore xyz > (x+z-y)(x+y-z)(z+y-x)$

b) i $\frac{1}{\cos x} = \frac{A \cos x}{1-\sin x} + \frac{B \cos x}{1+\sin x}$

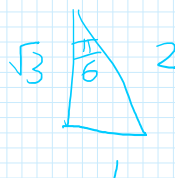
$$\frac{\cos^2 x}{\cos x} = A \cos x (1+\sin x) + B \cos x (1-\sin x)$$

$$1 = A(1+\sin x) + B(1-\sin x)$$

$$\therefore A+B=1$$

when $x = \frac{\pi}{2}$ $1 = A(1+1) + B(1-1)$
 $2A=1 \quad \therefore A=B=\frac{1}{2}$

ii $\int_0^{\frac{\pi}{6}} \sec x dx = \int_0^{\frac{\pi}{6}} \frac{\frac{1}{2} \cos x}{1-\sin x} + \frac{\frac{1}{2} \cos x}{1+\sin x}$
 $= \frac{1}{2} \int_0^{\frac{\pi}{6}} \frac{\cos x}{1+\sin x} - \frac{-\cos x}{1-\sin x}$
 $= \frac{1}{2} \left[\ln |1+\sin x| - \ln |1-\sin x| \right]_0^{\frac{\pi}{6}}$
 $= \frac{1}{2} \left[\left(\ln \left| 1+\frac{1}{2} \right| - \ln \left| 1-\frac{1}{2} \right| \right) - (\ln 1 - \ln 1) \right]$
 $= \frac{1}{2} \left[\ln \frac{3}{2} - \ln \frac{1}{2} \right]$



$$= \frac{1}{2} \left[\ln \frac{3}{2} - \ln \frac{1}{2} \right]$$

$$= \frac{1}{2} \ln 3$$

$$c) |t| = \sqrt{1+9+4} = \sqrt{14}$$

$$|u| = \sqrt{16+1+4} = \sqrt{21}$$

$$|v| = \sqrt{25+4+49} = \sqrt{78}$$

For the three sides to form a triangle,

$$|t| + |u| \geq |v|$$

$$\text{but } |t| + |u| = \sqrt{14} + \sqrt{21}$$

$$= 8.324$$

$$= \sqrt{69.292}$$

$$< \sqrt{78}$$

$\therefore$ so they cannot form a triangle.

$$d) \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx = \int_0^{\pi} \frac{(\pi-x) \sin x}{1+\cos^2 x} dx$$

$$= \pi \int_0^{\pi} \frac{\sin x}{1+\cos^2 x} dx - \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx$$

$f(x)$	$f(\pi-x)$
x	$\pi-x$
$\sin x$	$\sin(\pi-x) = \sin x$
$\cos^2 x$	$\cos^2(\pi-x) = \cos^2 x$

$$\therefore 2 \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx = \pi \int_0^{\pi} \frac{\sin x}{1+\cos^2 x} dx$$

$$\therefore \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx = \frac{\pi}{2} \int_0^{\pi} \frac{-\sin x}{1+\cos^2 x} dx$$

$$= \frac{-\pi}{2} \left[\tan^{-1} \cos x \right]_0^{\pi}$$

$$= \frac{-\pi}{2} \left[-\frac{\pi}{4} - \frac{\pi}{4} \right]$$

$$= \frac{\pi^2}{4}$$

e) i) Jimin didn't claim that $f(n)$ produced all prime numbers, just that all numbers produced are prime.

ii) $f(41) = 41^2 + 41 + 41$, which clearly has 41 as a factor.

$$u = (\ln x)^n \quad v = \frac{1}{2}x^2$$

$$u' = n \times \frac{1}{x} \times (\ln x)^{n-1} \quad v' = x$$

a) i $I_n = \int x (\ln x)^n dx$

$$= uv - \int u'v dx$$

$$= \frac{1}{2}x^2 (\ln x)^n - \int \frac{1}{2}x^2 \times \frac{1}{x} \times n \times (\ln x)^{n-1} dx$$

$$= \frac{1}{2}x^2 (\ln x)^n - \frac{1}{2}n \int x (\ln x)^{n-1} dx$$

$$= \frac{1}{2}x^2 (\ln x)^n - \frac{1}{2}n I_{n-1}$$

ii $I_0 = \int_1^3 x (\ln x)^0 dx$

$$= \left[\frac{x^2}{2} \right]_1^3$$

$$= \frac{9}{2} - \frac{1}{2}$$

$$= 4$$

$$I_1 = \int_1^3 x \ln x dx$$

$$= \left[\frac{1}{2}x^2 \ln x \right]_1^3 - \frac{1}{2}I_0$$

$$= \frac{1}{2} [9 \ln 3 - \ln 1] - \frac{1}{2} \times 4$$

$$= \frac{9}{2} \ln 3 - 2$$

$$I_2 = \int_1^3 x (\ln x)^2 dx$$

$$= \left[\frac{1}{2}x^2 (\ln x)^2 \right]_1^3 - \frac{1}{2} \times 2 \times I_1$$

$$= \frac{1}{2} [9 (\ln 3)^2] - \left[\frac{9}{2} \ln 3 - 2 \right]$$

$$= \frac{9}{2} (\ln 3)^2 - \frac{9}{2} \ln 3 + 2$$

b) i $r_1 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} + \lambda_1 \begin{bmatrix} 2 \\ 2 \\ -3 \end{bmatrix}$ $r_2 = \begin{bmatrix} 4 \\ -2 \\ 9 \end{bmatrix} + \lambda_2 \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$

$$x_1 = 1 + 2\lambda_1 \quad (1)$$

$$y_1 = 2\lambda_1 \quad (2)$$

$$z_1 = 2 - 3\lambda_1 \quad (3)$$

$$x_2 = 4 + \lambda_2 \quad (4)$$

$$y_2 = -2 + 2\lambda_2 \quad (5)$$

$$y_3 = 9 - 2\lambda_2 \quad (6)$$

$$(1) = (4)$$

$$1 + 2\lambda_1 = 4 + \lambda_2 \quad (1A)$$

$$(2) = (5)$$

$$2\lambda_1 = -2 + 2\lambda_2 \quad (2A)$$

sub (2A) into (1A)

$$1 + (-2 + 2\lambda_2) = 4 + \lambda_2$$

$$-1 + 2\lambda_2 = 4 + \lambda_2$$

$$\lambda_2 = 5$$

$$\text{when } \lambda_2 = 5, \quad 2\lambda_1 = -2 + 2(5) \\ = 8 \\ \therefore \lambda_1 = 4$$

$$\text{when } \lambda_1 = 4, \lambda_2 = 5$$

$$x_1 = 9$$

$$y_1 = 8$$

$$z_1 = -10$$

$$x_2 = 9$$

$$y_2 = 8$$

$$z_2 = -1$$

$\therefore$ they do not intersect

ii

$$\begin{aligned} d &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \\ &= \sqrt{(4 + \lambda_2 - 1 - 2\lambda_1)^2 + (-2 + 2\lambda_2 - 2\lambda_1)^2 + (9 - 2\lambda_2 - 2 + 3\lambda_1)^2} \\ &= \sqrt{(3 + \lambda_2 - 2\lambda_1)^2 + (-2 + 2\lambda_2 - 2\lambda_1)^2 + (7 - 2\lambda_2 + 3\lambda_1)^2} \\ &= \sqrt{\begin{aligned} &3^2 + \lambda_2^2 + 4\lambda_1^2 + 2(3\lambda_2 - 6\lambda_1 - 2\lambda_1\lambda_2) \\ &+ 4 + 4\lambda_2^2 + 4\lambda_1^2 + 2(-4\lambda_2 + 4\lambda_1 - 4\lambda_1\lambda_2) \\ &+ 49 + 4\lambda_2^2 + 9\lambda_1^2 + 2(-14\lambda_2 + 21\lambda_1 - 6\lambda_1\lambda_2) \end{aligned}} \\ &= \sqrt{9\lambda_2^2 + 17\lambda_1^2 + 62 + \lambda_1(-12 + 8 + 42) + \lambda_2(6 - 8 - 28) + \lambda_1\lambda_2(-4 - 8 - 12)} \\ &= \sqrt{9\lambda_2^2 + 17\lambda_1^2 + 62 + 38\lambda_1 - 30\lambda_2 - 24\lambda_1\lambda_2} \\ &= \sqrt{9\lambda_2^2 + 16\lambda_1^2 + 40\lambda_1 - 30\lambda_2 - 24\lambda_1\lambda_2 + 25 + (\lambda_1^2 - 2\lambda_1 + 1) + 36} \\ &= \sqrt{9\lambda_2^2 + 16\lambda_1^2 + 25 - 2(12\lambda_1\lambda_2 + 15\lambda_2 - 20\lambda_1) + (\lambda_1 - 1)^2 + 36} \\ &= \sqrt{(3\lambda_2 - 4\lambda_1 - 5)^2 + (\lambda_1 - 1)^2 + 36} \end{aligned}$$

* by the result given.

iii Minimum distance when $(\lambda_1 - 1) = 0$ and $(3\lambda_2 - 4\lambda_1 - 5) = 0$

$$\therefore d = \sqrt{0^2 + 0^2 + 36} \\ = 6$$

iv

$$\lambda_1 - 1 = 0$$

$$\therefore \lambda_1 = 1$$

$$3\lambda_2 - 4 - 5 = 0$$

$$3\lambda_2 = 9$$

$$\lambda_2 = 3$$

$$\text{when } \lambda_1 = 1, \begin{pmatrix} 1+2 \\ 0+2 \\ 2-3 \end{pmatrix} \Rightarrow (3, 2, -1)$$

$$\text{when } \lambda_2 = 3, \begin{pmatrix} 4+3 \\ -2+6 \\ 9-6 \end{pmatrix} \Rightarrow (7, 4, 3)$$

Question 16

a) The proof will be by contrapositive

if r is irrational $\Rightarrow \sqrt{r}$ is irrational

CONTRAPOSITIVE: if $\sqrt{r}$ is rational, then r is rational

if $\sqrt{r}$ is rational, then let $\sqrt{r} = \frac{a}{b}$ $a, b \in \mathbb{Z}$

$$\therefore r = \frac{a^2}{b^2}$$

$\therefore r$ is also rational

$\therefore$ if $\sqrt{r}$ is rational, then r is rational

$\therefore$ if r is irrational, then $\sqrt{r}$ is also irrational

b) i^i

$$\text{let } i = e^{i\frac{\pi}{2}}$$

$$\therefore i^i = (e^{i\frac{\pi}{2}})^i$$

$$= e^{i^2\frac{\pi}{2}}$$

$$= e^{-\frac{\pi}{2}}$$

$$\approx 0.207$$

which is real.

c) $t = \tan \frac{x}{2}$

i $\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$

$$\therefore \frac{dx}{dt} = \frac{2}{\sec^2 \frac{x}{2}}$$

$$= \frac{2}{1 + \tan^2 \frac{x}{2}}$$

$$= \frac{2}{1 + t^2}$$

ii $\frac{2 \tan \frac{x}{2}}{1 + \tan^2 \frac{x}{2}} = \frac{2 \tan \frac{x}{2}}{\sec^2 \frac{x}{2}}$ $\times \cos^2 \frac{x}{2}$ $\times \cos^2 \frac{x}{2}$

$$= 2 \tan \frac{x}{2} \times \cos^2 \frac{x}{2}$$

$$= 2 \frac{\sin \frac{x}{2}}{\cos \frac{x}{2}} \times \cos^2 \frac{x}{2}$$

$$= 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

$$= \sin \left(\frac{x}{2} + \frac{x}{2} \right)$$

$$= \sin x$$

iii $\int_0^{\frac{\pi}{2}} \frac{1}{1 + k \sin x} dx$

$$= \int_0^1 \frac{1}{1 + \left(\frac{2kt}{1+t^2} \right)} \cdot \frac{2}{1+t^2} dt$$

let $t = \tan \frac{x}{2}$

$$\therefore \frac{dx}{dt} = \frac{2}{1+t^2}$$

$$dx = \frac{2dt}{1+t^2}$$

$$k \sin x = \frac{2kt}{1+t^2}$$

$$\int \frac{f'(x)}{a^2 + [f(x)]^2} dx = \frac{1}{a} \tan^{-1} \frac{f(x)}{a}$$

$$t = \tan \frac{x}{2}$$

$$\frac{dx}{dt} = \frac{2}{1+t^2}$$

$$\sin x = \frac{2t}{1+t^2}$$

$$\int_0^1 \frac{1}{(1+t^2)} \times \frac{1}{1+t^2} dt$$

$$= 2 \int_0^1 \frac{1}{t^2 + 2kt + 1} dt$$

$$= 2 \int_0^1 \frac{1}{t^2 + 2kt + k^2 + 1 - k^2} dt$$

$$= 2 \int_0^1 \frac{1}{(t+k)^2 + (1-k^2)} dt$$

$$= 2 \left[\frac{1}{\sqrt{1-k^2}} \tan^{-1} \left(\frac{t+k}{\sqrt{1-k^2}} \right) \right]_0^1$$

$$= \frac{2}{\sqrt{1-k^2}} \left[\tan^{-1} \frac{1+k}{\sqrt{1-k^2}} - \tan^{-1} \frac{k}{\sqrt{1-k^2}} \right]$$

$$= \frac{2}{\sqrt{1-k^2}} \tan^{-1} \frac{\sqrt{1-k}}{\sqrt{1+k}}$$

$$= \frac{2}{\sqrt{1-k^2}} \tan^{-1} \sqrt{\frac{1-k}{1+k}}$$

iv $I_{n+1} + 2I_n = \int_0^{\frac{\pi}{2}} \frac{\sin^{n+1} x}{2 + \sin x} dx + 2 \int_0^{\frac{\pi}{2}} \frac{\sin^n x}{2 + \sin x} dx$

$$= \int_0^{\frac{\pi}{2}} \frac{\sin^{n+1} x + 2 \sin^n x}{2 + \sin x} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{\sin^n x (\sin x + 2)}{2 + \sin x} dx$$

$$= \int_0^{\frac{\pi}{2}} \sin^n x dx$$

v let $n=1$
 $I_2 + 2I_1 = \int_0^{\frac{\pi}{2}} \sin x dx$

$$\therefore I_2 = \int_0^{\frac{\pi}{2}} \sin x dx - 2I_1$$

$$k \sin x = \frac{2kt}{1+t^2}$$

when $x=0$, $t=0$
 $x=\frac{\pi}{2}$, $t=1$

$$\sin x = \frac{t}{1+t^2}$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

$$(A-B) = \tan^{-1} \left(\frac{\tan A - \tan B}{1 + \tan A \tan B} \right)$$

$$\text{let } x = \tan^{-1} \frac{1+k}{\sqrt{1-k^2}} - \tan^{-1} \frac{k}{\sqrt{1-k^2}}$$

$$\tan x = \tan \left(\tan^{-1} \frac{1+k}{\sqrt{1-k^2}} - \tan^{-1} \frac{k}{\sqrt{1-k^2}} \right)$$

$$= \frac{\frac{1+k}{\sqrt{1-k^2}} - \frac{k}{\sqrt{1-k^2}}}{1 + \frac{1+k}{\sqrt{1-k^2}} \times \frac{k}{\sqrt{1-k^2}}}$$

$$= \frac{1}{\sqrt{1-k^2}}$$

$$1 + \frac{k+k^2}{1-k^2}$$

$$= \frac{1}{\sqrt{1-k^2}} \times (1-k^2)$$

$$= \frac{1-k^2+k+k^2}{1-k^2} \times (1-k^2)$$

$$= \frac{\sqrt{1-k^2}}{1+k}$$

$$= \frac{\sqrt{1-k} \sqrt{1+k}}{1+k}$$

$$= \frac{\sqrt{1-k}}{\sqrt{1+k}}$$

$$\therefore x = \tan^{-1} \frac{\sqrt{1-k}}{\sqrt{1+k}}$$

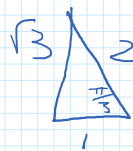
$$T = 1 \rightarrow \left[\frac{\pi}{2} \right] - \left[\frac{\pi}{2} \right]$$

$$\begin{aligned}
 \therefore I_2 &= \int_0^{\frac{\pi}{2}} \sin x - 2I_1 \\
 &= \left[-\cos x \right]_0^{\frac{\pi}{2}} - 2I_1 \\
 &= -[0 - 1] - 2I_1 \\
 &= 1 - 2I_1
 \end{aligned}$$

also when $n=0$

$$\begin{aligned}
 I_1 + 2I_0 &= \int_0^{\frac{\pi}{2}} 1 dx \\
 I_1 &= \left[x \right]_0^{\frac{\pi}{2}} - 2I_0 \\
 &= \frac{\pi}{2} - 2I_0
 \end{aligned}$$

$$\begin{aligned}
 I_0 &= \int_0^{\frac{\pi}{2}} \frac{(\sin x)^0}{2 + \sin x} dx \\
 &= \int_0^{\frac{\pi}{2}} \frac{1}{2 + \sin x} dx \\
 &= \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{1}{1 + \frac{1}{2} \sin x} dx \\
 &= \frac{1}{2} \times \frac{2}{\sqrt{1 - \frac{1}{4}}} \tan^{-1} \sqrt{\frac{\frac{1}{2}}{1 - \frac{1}{2}}} \\
 &= \frac{1}{\sqrt{\frac{3}{4}}} \tan^{-1} \sqrt{\frac{1}{3}} \\
 &= \frac{2}{\sqrt{3}} \sqrt{\frac{\pi}{6}} \\
 &= \frac{\pi}{3\sqrt{3}}
 \end{aligned}$$



$$\begin{aligned}
 I_2 &= 1 - 2 \left[\frac{\pi}{2} - 2 \left[\frac{\pi}{3\sqrt{3}} \right] \right] \\
 &= 1 - \pi + \frac{4\pi}{3\sqrt{3}} \\
 &= 1 - \frac{9\pi}{9} + \frac{4\sqrt{3}\pi}{9} \\
 &= \left(\frac{4\sqrt{3}-9}{9} \right) \pi + 1
 \end{aligned}$$